NexRay AI

Medical Assistant Platform • Clinical Decision Support Report

23 July 2026 | 10:22 PM

Session ID: 23 **Modules Used:** X-Ray Analysis **Date:** 23 July 2026 | 10:22 PM

X-Ray Analysis

Detected Region: Chest (Posteroanterior View)

Possible Condition	Confidence
Pleural Effusion (Right side)	82%
Pneumonia or Pulmonary Consolidation	68%
Congestive Heart Failure	55%

■ **Urgency Level: Urgent — Same day attention required**

Recommended Tests

- Chest ultrasound or CT thorax to characterize pleural effusion and assess underlying lung parenchyma
- Pleural aspiration and fluid analysis (cell count, protein, LDH, glucose, culture) if effusion is significant
- Echocardiography to assess cardiac function
- Blood cultures if infection suspected
- Complete blood count and biochemistry panel

Suggested Treatment

- Treat underlying cause (antibiotics for pneumonia, diuretics for heart failure, etc.)
- Supportive care with oxygen therapy if hypoxic
- Consider thoracentesis if effusion is causing respiratory compromise
- Monitor fluid intake and output
- Serial chest imaging to assess response to treatment

Next Steps

- Correlate clinical presentation with imaging findings
- Perform detailed physical examination including vital signs and assessment of respiratory status
- Order urgent pleural fluid analysis if clinical deterioration or significant respiratory symptoms
- Initiate empiric antibiotics if pneumonia suspected pending culture results
- Arrange echocardiography to assess cardiac contribution

Summary: *Chest X-ray demonstrates a significant right-sided pleural effusion with blunting of the right costophrenic angle and right lower lobe opacity. Findings are consistent with pleural effusion secondary to pneumonia, heart failure, or other systemic conditions requiring further clinical correlation and investigation.*